J. Halzoom's Coffee Grid

time limit per test: 1 second
memory limit per test: 256 megabytes

After securing enough Moonlit Moss to keep their lantern flickering, SAM07A, OMAR, and ABD-ELMOHAYMEN ventured deeper into a particularly ancient and silent part of the Unknown — the Elderwood. The trees were impossibly tall, their bark etched with glowing glyphs that shifted when no one looked directly at them.

Their journey came to a halt at a moss-covered stone monolith. As they approached, the surface shimmered and revealed a magical window into another part of the forest — a humble cabin hidden among the trees. Inside lived a curious engineer named HALZOOM, famed not only for his inventions but also for brewing the finest magical coffee in the Unknown.

In that cabin, HALZOOM faced a peculiar problem. His comical cat, 3ntar, had recently fallen in love with a mysterious, graceful feline who roamed the Elderwood. 3ntar, now struck by divine inspiration, demanded a coffee that matched his "love mood" — a special number k.

To prepare this enchanting blend, HALZOOM turned to his experimental grid of coffee beans. Each cell in the grid held a smell strength, and the aroma of the final cup must come from a rectangle (submatrix) of beans whose total smell equals exactly k.

The beautiful cat that **3ntar** fell in her love.

His challenge? Find all rectangles (submatrices) in the grid where the sum of values is exactly k. For every value inside any of those rectangles, keep it. All other values must be changed to 0.

Now, it's your task to help HALZOOM, 3ntar, and the Elderwood explorers all at once — by solving this aromatic mystery.

Input

The first line contains integers n, m, k $(1 \le n, m \le 100,\ 0 \le k \le 10^4)$ — the number of rows, columns, and the desired total smell value.

The next n lines contain m integers each — the smell values in the grid $(0 \le a_{ij} \le 100)$.

Output

Print the grid after finding all valid rectangles. Keep the values that are part of at least one rectangle with total sum k, and change all others to 0.

Example

input Copy

```
5 5 5
1 1 1 1 5
1 7 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 5
```

output Copy

```
1 0 1 1 5
1 0 1 1 0
1 1 1 1 1
1 1 1 1 1
1 0 1 1 5
```

Note

In the first example:

1	1	1	1	5
1	7	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	5

The highlight on columns and rows, those submatrices will be printed with the same values.